Provability plus inconsistency of the added formula yields inconsistency

The Open Logic Project contributors

openlogic:first-order-logic/sequent-calculus:local-snapshot-20261013; CC BY 4.0

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Open Logic Project unit file sources/olp-first-order-logic-sequent-calculus/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the derivation of the empty right side is written in the source with a trailing \\quad after the sequent arrow, delivered verbatim

Editorial note. The label prop:provability-contr names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own prop environment in the unit source.

Allowed standard prerequisites (editorial).

- Cut rule. From derivations of Gamma_0 sequent A and of A, Gamma_1 sequent (empty right side) one derives Gamma_0, Gamma_1 sequent (empty).. The displayed derivation cuts pi_0 (Gamma_0 sequent A) against pi_1 (A, Gamma_1 sequent empty).
- Inconsistency via derivable empty sequent. Gamma is inconsistent iff some finite Gamma' subseteq Gamma has Gamma' sequent (empty right side) derivable in LK.. The cut conclusion Gamma_0, Gamma_1 sequent (empty) with Gamma_0, Gamma_1 subseteq Gamma witnesses the inconsistency.

Proposition 0.1. *If $\Gamma \vdash A$ and $\Gamma \cup \{A\}$ is inconsistent, then Γ is inconsistent.*

Proof. There are finite Γ_0 and $\Gamma_1 \subseteq \Gamma$ such that **LK** derives $\Gamma_0 \Rightarrow A$ and $A, \Gamma_1 \Rightarrow$. Let the **LK**-derivation of $\Gamma_0 \Rightarrow A$ be π_0 and the **LK**-derivation of $\Gamma_1, A \Rightarrow$ be π_1. We can then derive

$$\dfrac{\begin{array}{cc}\vdots\,\pi_0 & \vdots\,\pi_1 \\ \Gamma_0 A & A, \Gamma_1\end{array}}{\Gamma_0, \Gamma_1}\ \text{Cut}$$

Since $\Gamma_0 \subseteq \Gamma$ and $\Gamma_1 \subseteq \Gamma$, $\Gamma_0 \cup \Gamma_1 \subseteq \Gamma$, hence Γ is inconsistent. $\square$